Airline Reservation System

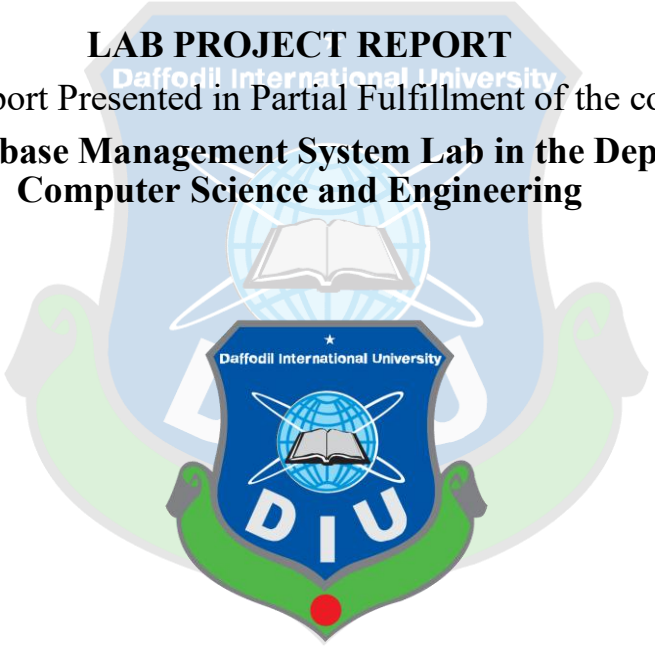
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LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course

CSE312: Database Management System Lab in the Department of Computer Science and Engineering



DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Md. Hasanuzzaman Dipu**, **Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Implement and optimize relational databases using basic to advanced SQL queries, indexing techniques, and query optimization strategies.
CO2	Analyze and apply security techniques, distributed database architectures, and emerging tools in database management to address real-world challenges.
CO3	Perform effectively as an individual or a member or a leader of diverse teams through proper documentation and initialization of project work.
CO4	Create a project by explaining complex computer engineering activities with the computer engineering community by performing effective communication through demonstration and presentation

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP	CEA
CO1	PO3	C3, C5, A2, P1, P2	K4-6	EP1, EP4	
CO2	PO5	C3, C4, A2, P2, P3	K4, K6, K8	EP2	
CO3	PO9	A1, A2, A4, P3			
CO 3	PO10	P2, A2, A4			EA1, EA2

The mapping justification of this table is provided in section **4.3.1**, **4.3.2** and **4.3.3**.

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Chapter 1

Introduction

1.1 Introduction

The flight reservation system is a user-friendly platform designed to simplify booking and managing air travel for passengers and airlines. Passengers, identified by unique Passenger IDs, provide essential details like name, date of birth, phone number, and address. The system supports multiple reservations, each tied to a unique Reservation ID and specific travel dates. Payments are recorded efficiently, capturing details like payment ID, amount, date, method, and card information, ensuring a seamless process.

Flights, the core of the system, are identified by unique Flight IDs and include information such as seating capacity, destination, and scheduled arrival and departure times. These flights are operated by airplanes, each with a unique Airplane ID, name, and capacity. Departures and arrivals occur at airports identified by Airport IDs, featuring names, locations, cities, and countries, facilitating smooth travel connections.

Reservations may include one or more tickets, each identified by a unique Ticket ID. Tickets specify seat numbers, prices, and travel classes (economy, business, or first class), offering passengers flexibility. The system connects passengers to flights, enabling them to book seats, select preferences, and manage reservations easily. Payment options support multiple methods for user convenience.

For airlines, the system streamlines operations by simplifying flight scheduling, capacity management, and passenger tracking. It enhances efficiency and ensures smooth communication between passengers and airlines. Overall, the system bridges the gap between users and airlines, making air travel more accessible, faster, and hassle-free.

1.2 Motivation

The motivation behind developing this project is to gain practical experience in database design and implementation. Airlines worldwide already use advanced reservation systems, but as students, building a simplified version helps us understand the core concepts of Entity-Relationship modeling, normalization, SQL queries, and database connectivity. It not only improves technical knowledge but also prepares us for real-world projects.

1.3 Objectives

- To design a normalized database for airline reservations
- To implement the system in Oracle 11g XE
- To demonstrate data insertion, queries, and DB connection
- Oracle SQL developer
- Ojdbc8.jar (Oracle JDBC Driver for Java to Oracle connection)
- NetBeans IDE (for Java frontend)

1.4 Feasibility Study

The system is technically feasible because:

- **Technical Feasibility:** Oracle 11g XE provides all required DBMS features (ER modeling, normalization, SQL queries, JDBC connectivity).
- **Operational Feasibility:** The system can be easily used by students, teachers, and small-scale organizations to demonstrate ticket booking.
- **Economic Feasibility:** It is low cost since it only requires Oracle XE (free) and Java NetBeans IDE (free).
- **Time Feasibility:** The project was completed within the semester timeline with group effort.

1.5 Gap Analysis

- **Existing Solutions:** Airlines already use global reservation systems like Amadeus, Galileo, Sabre, but those are large-scale commercial systems.
- **Gap Identified:** These systems are too complex for academic learning and are not open-source.
- **Proposed Solution:** Our project builds a simplified academic-level system focusing on DB design, normalization, and queries. This bridges the gap between theoretical learning and real-world applications.

1.6 Project Outcome

- Fully designed normalized schema.
- Implemented tables in Oracle with sample data.
- Queries tested (joins, views, decode functions).
- Successful DB connection via Java JDBC.

Chapter 2

System Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Requirements

The Airline Reservation System has the following **requirements**:

Functional Requirements

- A passenger must be able to create an account.
- A passenger can make multiple reservations.
- A reservation can include one or more tickets.
- Each ticket must have a seat number, class (Economy/Business/First), and price.
- Payments must be linked to reservations.
- A flight must have details such as FlightID, destination, capacity, arrival, and departure times.
- An airplane can operate multiple flights.
- Each flight must depart from and arrive at an airport.

Non-Functional Requirements

- **Performance:** Queries must run efficiently even with large data.
- **Reliability:** Data must be consistent and free from anomalies.
- **Security:** Passenger and payment data must be protected.
- **Scalability:** The system should handle growing number of flights and passengers.

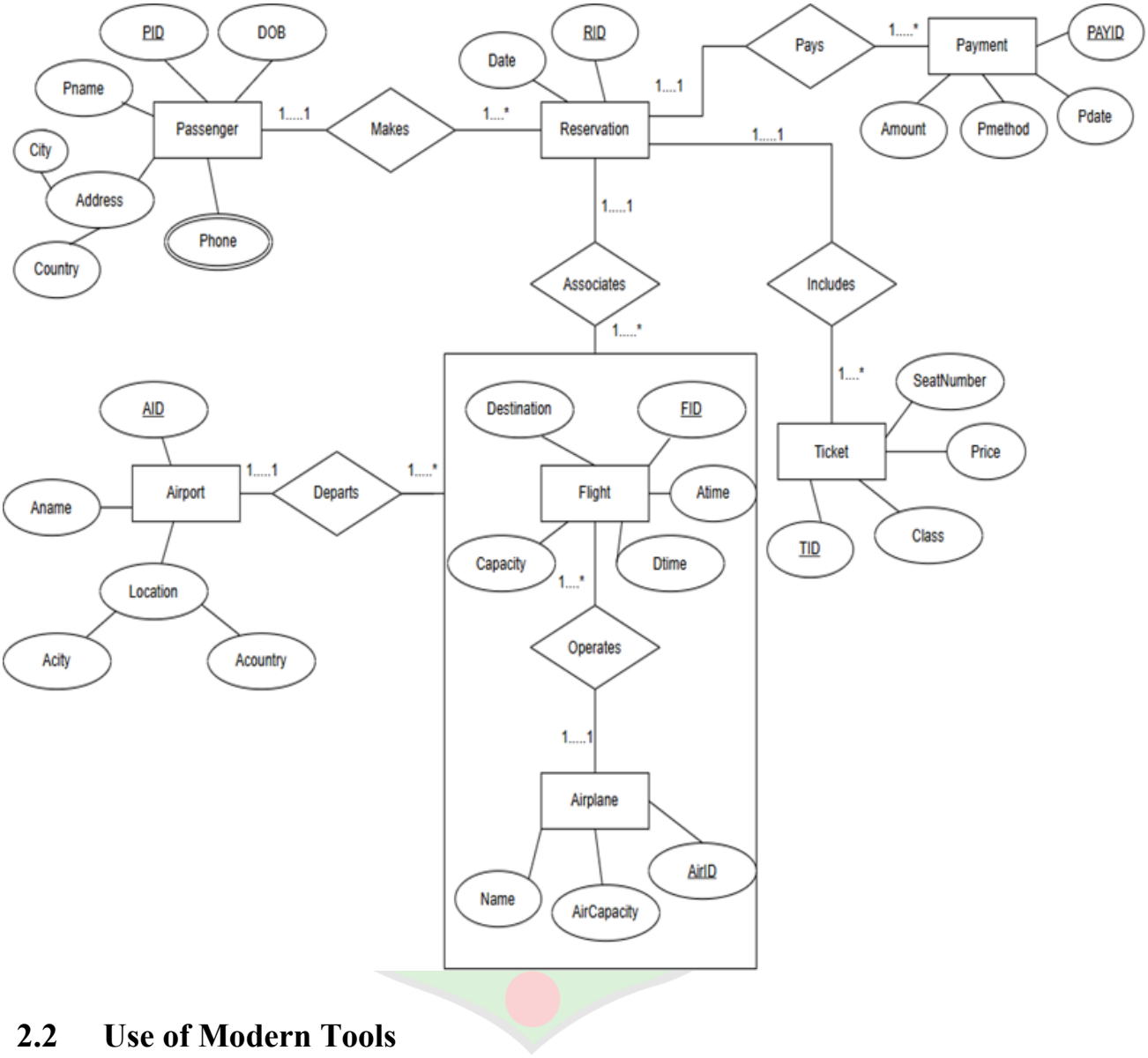
2.1.2 Entity Relationship Diagram

Based on the requirements, the following **design** was prepared:

- **Entities Identified:** Passenger, Reservation, Ticket, Payment, Flight, Airplane, Airport.

- **Relationships:**
  - ❖ Passenger **makes** Reservation
  - ❖ Reservation **includes** Ticket
  - ❖ Passenger **pays** Payment
  - ❖ Flight **operates** Airplane
  - ❖ Flight **departs/arrives** at Airport

ER Diagram:



2.2 Use of Modern Tools

For the implementation of the Airline Reservation System, the following modern tools were used:

- **Oracle 11g Express Edition (XE):**
  Used as the relational database management system (RDBMS). It provided the platform for creating tables, defining relationships, inserting data, and executing SQL queries.
- **Oracle SQL Developer:**
  A GUI tool that was used to interact with Oracle 11g XE. It allowed us to:
  - ❖ Write, execute, and test SQL queries easily.
  - ❖ Visualize the database schema.
  - ❖ Perform data insertion and validation.
  - ❖ Debug and optimize SQL commands.
- **NetBeans IDE:**

Used to establish the database connectivity (JDBC) with Oracle. In NetBeans, we:

- ❖ Integrated the ojdbc8.jar driver.
- ❖ Wrote Java programs to connect with Oracle.
- ❖ Tested queries through the Java frontend.
- ❖ Prepared the foundation for building a user-friendly GUI in the future.

Importance of Using Modern Tools:

- These tools ensured smooth development and debugging of the system.
- Provided a professional environment to simulate a real-world airline reservation system.
- Reduced complexity by providing GUI support for database operations and code integration.

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Chapter 3

Implementation and Results

3.1 Implementation

The database was implemented using Oracle 11g XE with the support of Oracle SQL Developer as the interface. The implementation followed the ERD designed earlier and was normalized to minimize redundancy.

3.1.1 Table Creation

The system consists of seven main entities:

- Passenger → stores details of travelers (PID, Name, Phone, DOB, City, Email, Gender, PassportNo, Nationality).
- Airport → contains airport information (Code, Name, City, Country).
- Airplane → stores airplane data (AID, Model, Capacity, Manufacturer, Last_Maintenance).
- Flight → represents a scheduled journey between two airports (FlightID, FromAirport, ToAirport, Departure, Arrival, AID, Base_Price, Status).
- Reservation → maintains booking records (RID, Date, Status, Total Amount).
- Reservation_Passenger → a junction table to support many-to-many relations between passengers and reservations.
- Ticket → represents actual seat bookings on a flight (TID, RID, FlightID, SeatNo, Class, Price, Status).
- Payment → records payment details linked to reservations (PayID, RID, Amount, Date, Method, TransactionID, Status).

3.1.2 Constraints and Integrity Rules

- Primary Keys ensure unique identification of records.
- Foreign Keys link entities (e.g., Reservation → Passenger, Flight → Airplane).
- CHECK constraints restrict values (e.g., Gender ∈ {Male, Female, Other}; Payment.Status ∈ {Completed, Failed, Refunded}).
- UNIQUE constraints prevent duplicate seat booking in the same flight.

3.1.3 Sequences and Triggers

Oracle does not support auto-increment columns by default. To handle this, sequences were created for Passenger, Reservation, Payment, Airplane, Flight, and Ticket. Triggers were written to auto-assign primary keys using these sequences during insertion.
This reduces human error and ensures smooth insertion of new records.

3.1.4 Indexes

Indexes were created on frequently searched attributes:

- idx_flight_departure (for flight schedules).
- idx_flight_route (for searching flights between airports).
- idx_passenger_name, idx_passenger_phone (for faster passenger lookups).
- idx_payment_status, idx_payment_date (for financial reporting).

These indexes improve query performance significantly when the database grows.

3.1.5 Sample Data Insertion

Data was inserted into all tables for testing. For example:

- Airports → JFK, LAX, DXB, LHR, HND, SYD, DEL, SIN.
- Airplanes → Boeing 737, Airbus A320, Boeing 787, Airbus A380.
- Flights → Routes such as JFK–LAX, LAX–JFK, JFK–LHR, HND–DXB.
- Passengers → John Smith, Sarah Johnson, Mohammed Al-Farsi, Priya Patel.

3.2 Output

To validate the database design, several verification queries were executed.

3.2.1 Record Counts

```
SELECT 'Airports' AS Table_Name, COUNT(*) AS Record_Count FROM Airport
UNION ALL
SELECT 'Airplanes', COUNT(*) FROM Airplane
UNION ALL
SELECT 'Flights', COUNT(*) FROM Flight
UNION ALL
SELECT 'Passengers', COUNT(*) FROM Passenger;
```

Output:

Table_Name	Record_Count
Airports	8
Airplanes	4
Flights	6
Passengers	4

This confirms that initial sample data was inserted successfully.

3.2.2 Passenger Reservation Analysis

```
SELECT DISTINCT
p.PID as "Passenger_Id",
p.Pname AS "Passenger_Name",
```

```
p.Phone AS "Phone",
p.Email AS "Email",
COUNT(r.RID) AS "Total_Bookings"
FROM Passenger p
JOIN Reservation_Passenger rp ON p.PID = rp.PID
JOIN Reservation r ON rp.RID = r.RID
WHERE r.Status = 'Confirmed'
GROUP BY p.PID, p.Pname, p.Phone, p.Email
ORDER BY p.PID;
```

Output :

Passenger_Id	Passenger_Name	Phone	Email	Total_Bookings
1	John Smith	1234567890	john.smith@email.com	2
2	Sarah Johnson	9876543210	sarah.j@email.com	1

This query demonstrates the system's ability to track customer bookings efficiently.

3.3 Discussion

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The implementation demonstrates the following:

- Normalization → Avoids redundancy (e.g., airport info stored only once).
- Triggers and Sequences → Ensure automation and reduce manual errors.
- Indexes → Enhance performance for queries involving flights and passengers.
- Constraints → Guarantee data consistency (e.g., no invalid gender or payment method).
- Scalability → The schema supports addition of more airports, flights, and passengers without modification.

Overall, the results confirm that the designed system meets the requirements of a simplified yet realistic airline reservation system.

Chapter 4

Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

The Airline Reservation System directly improves the **quality of life** for passengers and airline staff. Passengers can book, modify, and cancel tickets from anywhere, reducing the stress of long queues at counters. The system also minimizes errors in booking details, ensuring passengers have a smoother travel experience.
For employees, the system reduces workload by automating manual processes such as reservation management, payment tracking, and flight schedules. This creates a more **reliable, faster, and user-friendly** travel ecosystem, ultimately saving time, reducing errors, and improving convenience in daily life.

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4.1.2 Impact on Society & Environment

At a societal level, the system improves the efficiency of air travel by making reservation processes faster and more accurate, thereby reducing delays and mismanagement of flights. This fosters better connectivity across regions and supports globalization and tourism.
From an environmental perspective, the system promotes **digital ticketing**, reducing the dependency on paper-based tickets and receipts. This shift to paperless transactions helps in **sustainable resource usage**, lowering paper waste and contributing to environmental conservation. Moreover, optimized scheduling and capacity management reduce unnecessary fuel usage, indirectly supporting greener aviation practices.

4.1.3 Ethical Aspects

The project carefully considers ethical principles in data handling and system design. Since sensitive information such as passenger details and partial payment information is stored, the system ensures **data privacy and security** through constraints, masked storage (only last 4 digits of card numbers), and access control measures. Furthermore, the project emphasizes **fairness and transparency** by documenting all design steps and query processes, ensuring accountability. Ethical practices in database design prevent unauthorized access, misuse of passenger data, or manipulation of bookings. This responsible handling of data builds **trust** between the airline and its customers, which is essential for any modern digital system.data.

4.1.4 Sustainability Plan

The system is scalable, modular, and eco-friendly. Paperless operations, indexing for

efficiency, and secure data management support long-term technical, environmental, and social sustainability.

4.2 Complex Engineering Problem

4.2.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

POs	Justification of Mapping (Project Perspective)
PO3	This project involves designing and implementing a relational database with efficient SQL queries, indexing, and optimization techniques, thereby addressing complex problem-solving and system design challenges (PO3).
PO5	The project incorporates modern database technologies such as RDBMS for handling data and ensuring application of contemporary tools and practices in real-world scenarios (PO5).
PO9	Through collaborative project work, every members were engaged in teamwork, task allocation, and documentation. (PO9).
PO10	The project requires present the developed database system based on the requirement gathered, prepare technical reports, and communicate ideas clearly. (PO10).

4.2.2 Complex Problem Solving

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	**EP2** Range of Conflicting Requirements	**EP3** Depth of Analysis	**EP4** Familiarity of Issues	**EP5** Extent of Applicable Codes	**EP6** Extent Of Stakeholder Involvement	**EP7** Inter-dependence
✓	✓		✓			

EP1: This project requires us to design, implement, and optimize a relational database system. These tasks involve tackling complex engineering challenges such as efficient query design, indexing strategies, and performance optimization, enabling us to develop strong problem-solving skills.

EP2: In this project, we utilize relational database management systems (RDBMS) and advanced SQL techniques to build and manage the database. Applying these fundamental tools in a practical environment strengthens our skills in using established engineering tools effectively.

EP4: We critically evaluate our database design and implementation by analyzing query performance, tuning indexing methods, and optimizing system throughput. This ongoing evaluation helps us refine the solution to meet quality and efficiency standards.

4.2.3 Complex Engineering Activities

Table 4.2: Mapping with complex engineering activities.

EA1	EA2	EA3	EA4	EA5
✓	✓			

EA1: In this DBMS project, we use systematic methods to gather all project requirements from users and stakeholders. We document the database schemas, query designs, and system workflows carefully in reports and slide decks. This preparation helps us organize the technical information logically and tailor it to the users' and audience's needs, ensuring clear understanding of the project's structure and functionality.

EA2: For this project, we use oral and visual presentation techniques to explain complex database concepts such as normalization, query optimization, and transaction management. By presenting the DBMS project clearly, we help users and evaluators understand the design decisions and technical details. This also allows us to improve our verbal communication skills, use visual aids effectively, and respond confidently to any questions from the audience.

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Chapter 5

Conclusion

5.1 Summary

The project successfully designed and implemented a **airline reservation system**. It covered database design, normalization, table creation, triggers, sequences, indexing, and query execution. Sample data validated the correctness of the design, and queries demonstrated passenger and reservation analysis.

5.2 Limitation

- Limited dataset (few airports, passengers, flights).
- No real-time front-end GUI integration.
- No stored procedures for advanced business rules.
- Does not yet address concurrency control in a multi-user environment.

5.3 Future Work

- Development of a web-based or mobile GUI.
- Integration with real airline APIs (flight status, live prices).
- Implementation of stored procedures and triggers for advanced automation.
- Adding role-based authentication for admins, passengers, and staff.
- Expanding dataset to simulate real-world airline scale.

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